

Alberto Hernández Espinosa
Mexico City, Mexico • +52 552 383 4438 • albertohernandezespinosa@gmail.com
<https://albertoHdzE.github.io/website/> • <https://linkedin.com/in/alberto-hernandeze> •
<https://github.com/albertoHdzE>

1 Professional Summary

- AI researcher with over 10 years of experience leading scientific software development for universities and industry, specializing in artificial intelligence, cognitive science, and complexity science.
- Expertise in designing and deploying machine learning, deep learning, and causal AI models, with a proven track record of delivering impactful solutions (e.g., 53% cloud cost reduction).
- Published with Oxford University and Oxford Immune Algorithmics, contributing to advancements in artificial intelligence, integrated information theory and algorithmic complexity.
- Skilled in remote collaboration, having worked globally for 7 years, and prepared to work with interdisciplinary AI teams.
- AI Engineer Lead at K-Square Group, driving AI integration and fostering innovation through agentic platforms and research.

2 Skills & Expertise

- AI/ML Technologies: Python, TensorFlow, PyTorch, Scikit-learn, Keras, LangGraph, Ollama, RAG, Agents, Small Model Tuning, Multi-Agent Orchestration, Tool-Use LLMs, Chain-of-Thought Prompting, Self-Reflection LLMs, LLM-as-a-Judge Patterns
- Programming Languages: Python, R, C++, SQL, Mathematica, FastAPI, React
- Cloud & DevOps: GCP, AWS, Docker, FastAPI
- Data Science: Big Data, SQL/NoSQL, ETL, Pandas, NumPy, Matplotlib
- Domains: Artificial Intelligence, Complexity Science, Cognitive Science, Reinforcement Learning, Algorithmic Information Theory, Enhancing RAG with Complexity, Theoretical Limits on Intelligence of LLMs, Agentic LLM Reasoning, Emergent Collective Intelligence

3 Professional Experience

3.1 AI Engineer Lead, K-Square Group (Remote)

Nov 2024–Present

- Conduct and lead teams for implementation of AI-driven projects, enhancing team collaboration and project execution.
- Spearhead research and development of AI/ML applications tailored to specific industry challenges, focusing on innovation and efficiency.
- Design and oversee the integration of advanced AI technologies, including agentic systems and complex data processing pipelines.
- Develop and implement strategic frameworks for rapid prototyping and proof-of-concept development, aligning with organizational goals.
- Foster a culture of experimentation and continuous improvement, mentoring teams in leveraging AI tools effectively.

3.2 Chief AI Scientist, MilkStraw AI, San Francisco, CA (Remote)

Oct 2022–Sep 2024

- Led research of AI-driven cloud resource optimization models on AWS, achieving up to 53% cost savings for clients.
- Designed and deployed ML pipelines using Python, TensorFlow, and PyTorch.
- Collaborated remotely with global teams to deliver scalable AI solutions.

3.3 Research Director, Laxford Capital (Remote)

Jun 2019–Jul 2022

- Directed AI trading initiatives, developing reinforcement learning agents for financial markets.
- Oversaw model design and implementation in Python and PyTorch.
- Enhanced trading performance through causal AI approaches.

3.4 Research Collaborator, Oxford Immune Algorithmics & Oxford University, Oxford, UK (Remote with Visits)

2015–Nov 2024

- Contributed to publications on integrated information theory and algorithmic complexity.
- Implemented models in Mathematica and Python.
- Co-authored papers with Oxford researchers, advancing cognitive science applications.

4 Education

4.1 PhD in Philosophy of Science, UNAM, Mexico City, Mexico

2015–2019 (Specialization: Artificial Intelligence and Cognitive science)

- Research focused on integrated information theory (Phi).
- CONACyT Scholarship.
- Education in association with University of Oxford, Computer Science Faculty.

4.2 Master in Computer Sciences, UNAM, Mexico City, Mexico

2012–2014 (Specialization: Artificial Intelligence)

- Thesis on computational creativity models.
- CONACyT Scholarship.

5 Selected Projects

- GraphRAG From Scratch
 - Clean, mathematically explicit implementation of RAG, RAU, and GraphRAG for scientific comparison.
 - Exposes embeddings, graph construction, traversal, and synthesis as separate components to study when graph-based retrieval improves multi-hop reasoning.
 - Code: <https://github.com/albertoHdzE/GraphRAG-scratch>
- HiFi
 - High-fidelity experimentation environment for AI systems, focused on reproducible workflows and controlled evaluation.
 - Organizes prototypes and implementation patterns for building and testing agentic applications.
 - Code: <https://github.com/albertoHdzE/HiFi>
- CouncilAI V2 (AI POCs Feature Spanner)
 - Reproducible benchmarking harness and dashboard for comparing vector, graph, long-context, and managed-search retrieval substrates under a fixed agentic workflow.
 - Measures latency, verifiability, and retrieval quality across multiple RAG pipelines using deterministic run artifacts and a React dashboard.
 - Code: <https://github.com/albertoHdzE/ai-pocs-feature-spanner>
- ContextForge (k-mcp-farm)
 - Open-source registry and proxy that federates MCP, A2A, and REST/gRPC APIs with centralized governance, discovery, and observability.
 - Consolidates tools, agents, and APIs behind a clean endpoint for AI clients, with plugin extensibility and Kubernetes-ready deployment.
 - Code: <https://github.com/albertoHdzE/k-mcp-farm>
- AG-UI
 - Pluggable agent-user interaction layer for human-machine communication in agentic applications.
 - Supports interactive interfaces that make complex AI workflows more observable and usable.
 - Code: <https://github.com/albertoHdzE/agui>
- CausalBool: Deterministic Causal Integration
 - Physics-inspired algorithmic information theory for Boolean networks, deriving exact closed-form formulae for causal reconstruction without probabilistic assumptions.
 - Implemented deterministic index algebra and ordering-aware invariances to scale information signatures (PID, Transfer Entropy) beyond combinatorial limits.
 - Code: <https://github.com/albertoHdzE/CausalBool>
- Noesis: Agentic Agnostic Platform
 - Platform for creating AI/ML models (regression, classification, clustering, anomaly detection) using LangGraph, Ollama, local tuned models, FastAPI, and React.
 - Applied to Fraud Detection in financial and insurance institutions.
 - Code: <https://github.com/KSProjectX/test-xx> (restricted access).
- KS-Onboarding: Agentic Platform for Project Control and Onboarding

- Deep agents for organizing projects, analyzing documentation/meetings, generating insights, and responding to questions using local/external information, OCR, and voice recognition.
- Advanced RAG enhanced with complexity science based on high-level research (<https://arxiv.org/abs/2505.02581>).
- Code: <https://github.com/KSProjectX/KS-onboarding> (restricted access).
- Agents for Automation of Running Tests
 - Agentic platform using user stories/use cases in natural language or code to dynamically create pytest test code and report coverage.
 - Code: <https://github.com/Sucharita8/KS-qe-platform> (restricted access).
- Stock Prices Prediction
 - Developed predictive and autonomous decision-making models for stock prices based on Deep Reinforcement Learning and advanced ML approaches.
 - Integrated time-series analysis and reinforcement learning.
 - Deployed on cloud platforms for real-time forecasting.
 - GitHub: https://github.com/albertoHdzE/finance_1, Detailed explanation: <https://sites.google.com/view/complexai/FINANCE>.
- MOSAICA: DNA Complexity Analysis
 - Investigated DNA sequence complexity using Machine learning with Python and Mathematica.
 - Correlated algorithmic complexity with DNA curvature.
 - Results published in academic papers.
 - GitHub: <https://github.com/albertoHdzE/MOSAICA>.

6 Publications

- 2025: “Neurodivergent Influenceability as a Contingent Solution to the AI Alignment Problem.” PNAS. <https://academic.oup.com/pnasnexus/article/5/4/pgag076/8651394>
- 2025: “SuperARC: An Agnostic Test for Narrow, General, and Super Intelligence Based On the Principles of Recursive Compression and Algorithmic Probability.” Nature. <https://www.nature.com/articles/s41467-026-73289-5>
- 2021: “Estimations of Integrated Information Based on Algorithmic Complexity and Dynamic Querying.” World Scientific. https://www.worldscientific.com/doi/10.1142/9789811235726_0005
- 2017: “Is There Any Real Substance to the Claims for a ‘New Computationalism’?” Springer. https://link.springer.com/chapter/10.1007/978-3-319-58741-7_2
- 2013: “Does the Principle of Computational Equivalence Overcome the Objections Against Computationalism?” Springer. https://link.springer.com/chapter/10.1007/978-3-642-37225-4_14

7 Certifications

- Mathematics for Machine Learning
- AI for Finance
- Google Cloud Specialization
- Reinforcement Learning

- Complexity Science

Full list of 47 certifications available here: <https://www.linkedin.com/in/alberto-hernandeze/details/certifications/>

8 Languages

- Spanish (mother language)
- English (advanced proficiency)
- German (Basic)
- French (Basic)
- Japanese (Basic)

Certifications available here: <https://www.linkedin.com/in/alberto-hernandeze/details/languages/>